Revenue-Contingent Financing Under Volatile Sales

Separating Price from Structure in a Paired Simulation Study

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Contents

What this study is, stated once up front. Reported simulated magnitudes are **simulation output under modelled assumptions**. No observed seller revenue, repayment, or default outcome exists anywhere in this study. The analytical propositions in §7 are **not** simulation output and are not covered by that qualifier.

Every quantitative statement carries a source note identifying its type and origin; the convention is set out in **Appendix A**, and the artifacts and checksums behind the simulated figures in **§15**.

The accompanying software at the address above is a **demonstration**. It holds no capital and makes no credit decisions. **As of August 2026, no external merchant data were used in this study; the public deployment is a demonstration, not a lending service.**

Abstract

Revenue-contingent financing makes each repayment a fixed share of realised revenue rather than a fixed instalment. The seller-side effect is mechanical: the payment falls when eligible revenue falls. The provider-side effect is conditional: recovery may **lead or lag** a cost-matched fixed schedule depending on the realised revenue path, by the exact condition of Proposition 4, and both directions occur in this scenario library. In the severe-downturn scenario it lags. The two effects are the same mechanism observed from opposite sides of the contract, and reporting either alone misstates the product — but the provider side must be reported with its scenario named, because it has no uniform direction.

This paper separates two questions that are routinely conflated: the **price** of a revenue-contingent contract, set by its cap factor, and its **structure**, meaning how payments re-

spond to revenue. Using a paired simulation in which a revenue-contingent contract and a cost-matched fixed instalment are applied to identical generated revenue paths, we find that at the illustrative cap factor `f = 1.20` the revenue-contingent arm removes **6.85** months above a burden threshold of 15% in a severe-downturn scenario, while recovering **65.46%** of its contractual target by month 12 against the fixed arm's **92.31% [S-1 | `baseline_v3_canonical.json → /scenarios/severe_downturn`]**. **These are simulated magnitudes, not observations.** The 15% threshold is an illustrative reporting band chosen for this study, not a validated hardship cutoff; burden is measured as payment ÷ GMV; and the fixed arm's recovery is *scheduled* recovery under the modelling assumption that every instalment is paid in full and on time, so it is an optimistic benchmark. Repricing at the nearest reference-path grid match `f* = 1.0945` changes the cost comparison entirely **[P-1, P-2 | `validation_v2_canonical.json → /pricing`]**. The **pre-cap payment rule is unchanged** — each payment remains the same fixed share of net sales — but the cap factor sets the contractual target and therefore moves completion timing, terminal clipping and the realised payment stream. Pricing and the revenue-contingent payment rule are **analytically separable**, yet **jointly determine realised outcomes**; which is why a cost ratio quoted at a single cap factor is a pricing result and not a property of revenue-contingent repayment as such.

We further characterise incomplete recovery exactly. Completion requires that some **finite** month `t ≤ H` satisfy `S_t ≥ f·A/r`; for a finite horizon this reduces to the terminal condition, but for an unbounded lifetime it does not, and equality of the lifetime sum is insufficient unless a finite partial sum attains the threshold. Where permanent revenue cessation precedes completion while a balance is outstanding, recovery genuinely fails: closure from month 7 leaves **100.0%** of simulated paths incomplete at both registered cap factors **[S-3, S-4 | `baseline_closure_v2_canonical.json` and `baseline_closure_equalcost_v2_canonical.json → /scenarios/closure_m7/RBF/incomplete_recovery_rate`]**. Incomplete recovery is not equivalent to principal loss, and we distinguish them.

We make no predictive, causal, affordability or default-prevention claim. The propositions are proved and hold for any revenue path; the magnitudes are illustrations under parameters we chose.

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1. Introduction and motivation

Access to finance is a persistent constraint on small enterprises in developing economies. The **2017 edition** of the IFC/SME Finance Forum *MSME Finance Gap* study estimated that approximately 40% of formal micro, small and medium enterprises in developing countries — some 65 million firms — **had** unmet financing needs, totalling $5.2 trillion annually (L-01). We cite that edition and figure in the past tense deliberately: the study has since been updated, and mixing editions would misreport both. That estimate is a benchmarking construct rather than an observation: it applies debt-to-sales ratios from ten developed markets to the enterprise population of each developing economy, and its own documentation states the assumption that firms "have the same willingness and ability to borrow as their counterparts in well-developed credit markets". We use it to motivate the question, not to parameterise anything.

The motivating application setting for this study is small e-commerce sellers in Vietnam, a policy-salient digital and e-commerce economy (L-02). **The availability of transaction records to a financier is a modelling premise of this study, not something L-02 or any other source we reviewed establishes.** We make no claim that Vietnamese platforms grant financiers access to seller revenue, returns, orders or fulfilment data, and no claim about seller-level observability in that market. **This study is Vietnam-motivated and illustratively parameterised, not Vietnam-calibrated.** No parameter in the specification was estimated from Vietnamese data, and we make no claim about the prevalence of thin credit-bureau files among Vietnamese sellers; we found no source that would support one.

Continuously recorded revenue makes a particular contract mechanically possible: repayment as a fixed percentage of revenue, continuing until a contractual cap is reached. The question this paper addresses is what that contract does — to the seller, and to the financier

— relative to a fixed instalment with the same contractual repayment target on the reference path, under the same revenue path.

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2. Why simulation, and what that costs

Answering the question empirically requires observing two contracts on the same seller over the same revenue realisation, and each seller takes at most one contract. **We did not identify a public dataset containing that counterfactual, nor a head-to-head comparison of seller burden and provider recovery under revenue-contingent versus cost-matched fixed schedules for small-business financing, in the searches documented through 2026-08-13** (Gap R-2; search protocol at `LITERATURE_MATRIX.md` §0.1). This question does not appear answered in the literature reviewed — a statement about our searches, not a claim that no such work exists.

A paired simulation supplies the missing counterfactual by construction: every contract is applied to the identical generated path, and all comparisons are within-path paired differences. The cost is equally clear. The revenue process is one we specified. The results describe **contract mechanics under stated assumptions**, and nothing about how sellers behave. We state this once here and again in §12, and we have tried not to blur it anywhere between.

Underwriting on cash-flow data — as distinct from repayment structure — is a separate and better-evidenced literature (L-04, L-05, L-06). We touch it only in §11, and only to disclaim.

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3. Related literature

Contingent repayment theory. The canonical treatment is the income-contingent loan literature, where repayment scales with realised income and the obligation carries consumption-smoothing and default-insurance properties (L-09). Nerlove (1975) states the standard critique — adverse selection, incentive effects, and the transfer of risk to the financier (L-10). This body is about **students**, not firms; the construct transfers, the findings do not, and we flag this at each use.

Revenue-based financing. Russel, Shi and Clarke (2025) is the closest published work to the contract studied here: transaction-level data from a South African payments platform, with repayment as automatic deduction of a fixed share of daily on-platform revenue until a set amount is repaid (L-07). They state directly that the structure "shifts risk to the financier, as repayment speed changes with revenue", and document that financed firms routed approximately 16% less revenue through the platform after eight months, slowing repayment. Two limits matter for us: it is a working paper, and its central finding is **behavioural revenue diversion**, which our model does not represent. Our under-reporting sweep is mechanical — we impose ω, we do not model a seller choosing it. Cordaro et al. (2022) report the first field experiment of a performance-contingent microfinance contract: working in **Kenya** with micro-distributors in a large food multinational's supply chain, they compare asset financing under a traditional debt contract against an equity-like contract, a hybrid debt–equity contract and an index-insurance contract, finding large positive impacts from the contractual innovations and improved sharing of risk and reward (L-08). None of those four designs is a percentage-of-revenue contract against a cap, so the mechanism differs from ours even though the motivation — leveraging improved observability of performance — is the same.

We did not identify a peer-reviewed academic study of the US merchant cash advance market in the searches documented through 2026-08-13 (Gap R-1). Those searches returned vendor marketing and search-optimised content, which the inclusion rules excluded. The authoritative US-market material we did identify is regulatory and legislative (L-23, L-24, L-26, L-45). We note the asymmetry — a product class with documented enforcement actions and recent legislation, without a corresponding academic literature in our results — but we do not treat the absence as an empirical finding about the field.

Repayment burden. Chapman and Dearden (2017) define repayment burden as the proportion of income required to service the obligation and argue it drives both default probability and living standards (L-11); Chapman and Lounkaew (2015) show fixed schedules concentrate burden severely in low-income states (L-12); Barr et al. (2019) state the design contrast plainly — fixed repayment creates high burden precisely when income is low, whereas income-linked repayment builds in automatic insurance against inability to repay (L-15). Chapman et al. (2010) establish the methodological point we adopt: burden must be computed **across the distribution**, not at the mean (L-13), which is why we report high-burden month counts and percentiles rather than means alone.

Two firm-side and household-side sources bear on the mechanism but test different instruments. Battaglia, Gulesci and Madestam (2024) randomise a payment-**deferral option** and find flexibility acts as insurance, raises risk-taking and lowers default (L-16) — this is not revenue-proportional repayment and we do not describe it as such. Ganong and Noel (2020) show that short-term payment size, not total obligation, drives default and consumption, since principal reduction had no effect while maturity extension had large ones (L-17) — consumer mortgages, and again a different instrument.

The price of contingency. Herbst and Hendren (2024) quantify the risk-shifting critique: adverse selection implies a typical borrower would repay about $1.64 in present value per dollar financed for an equity-like market to clear (L-18). Igan, Kim and Levy (2022) find persistent, material premia on state-contingent sovereign instruments (L-19). Neither is about SME finance. Both show that contingency **can** command a premium **in the settings studied** — where it transfers risk to the financier or creates adverse selection — but neither establishes a general rule, and we do not assert one for SME revenue-based financing. Whether such a premium arises there, and at what magnitude, is an open empirical question. What the two results do establish is that price and structure must be separated before either is discussed, since a contingent contract and a fixed one need not be priced alike for reasons that have nothing to do with the payment rule itself.

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4. Research question and contribution

Question. Under identical revenue paths, how do seller payment burden and provider recovery move together when repayment is revenue-contingent rather than fixed, holding the contractual repayment target constant on the reference path?

Contribution, in order of what we think carries weight:

1. **A paired, price-controlled comparison.** Principal, total contractual repayment and term are matched on a reference path, so the only difference between the primary arms is the *timing* of payments within the term — and the comparison is repeated at a second cap factor so that price is controlled rather than confounded.
2. **Two-sided measurement.** Seller payment burden and provider recovery are reported together for every scenario, because they are one mechanism seen from two sides and reporting either alone misstates the contract.
3. **Transparent failure and censoring boundaries.** The conditions under which recovery fails are characterised exactly, and the statistics that condition on completion are labelled as such rather than presented as portfolio outcomes.
4. **An analytical verification layer.** The propositions in §7 are not offered as novel theorems — most are elementary consequences of the contract definition. Their role is to establish which results are properties of the contract and which are artefacts of the simulation, and each is asserted numerically against the engine so the two cannot be confused.

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5. Contract definitions and comparison design

Let `B_t` be the remittance base in month `t` (contractually, net sales), `S_k = Σ_{t≤k} B_t`, `A` the advance, `f` the factor rate, `C = A·f` the contractual cap, and `r` the remittance rate.

Revenue-contingent arm (RBF). `p_t = min(r·B_t, C − Σ_{s<t} p_s)`, `p_t ≥ 0`. Payments continue until cumulative payments reach the cap.

Fixed arm, matched (FIX-A). `q_t = P` for `t ≤ N`, zero afterwards, with `N·P = C`. Principal, total contractual repayment and term are identical to RBF **on the reference path**. This isolates payment timing and is the primary comparison.

Fixed arm, external reference (FIX-B). An illustrative 18% nominal, 12-month amortizing schedule. The 18% is an assumption chosen for this study — **not a market rate**, and not observed or externally sourced anywhere in this project (spec amendment A-8).

At the illustrative cap factor `f = 1.20`, matching on the reference path gives a term of **13 months**, a payment of **17,076,923 VND** and an implied APR of **37.8694% [baseline_v3_canonical.json → /match_benchmark_a]**. At `f* = 1.0945` the matched term is **12 months**, payment **16,873,542 VND**, APR **18.3980% [baseline_equalcost_v2_canonical.json → /match_benchmark_a]**.

A comparability limit, stated wherever the arms appear together. An APR loan prices time; a factor-rate cap prices a multiple regardless of time. The two are not commensurable without conversion, and financial regulators have said so: New York's 23 NYCRR 600 requires standardised disclosure including APR for commercial financing under $2.5m (L-23), and California's disclosure regime identifies describing price as an "X% fee rate" or "Y% factor rate" — particularly where those diverge materially from APR — as a confusing representation (L-24). We report both, and we never quote one as the other.

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6. Simulation methodology

Simulation is an established method for comparing contingent against fixed repayment designs: Chapman and Lounkaew (2010) compare income-contingent and mortgage-type loan designs by simulation, and this study's paired design follows that precedent (L-14). The specification was frozen before any outcome analysis; amendments are logged with the result visible at the time. This matters because researcher degrees of freedom in choosing a data-generating mechanism and performance measures allow spurious claims of superiority for essentially any method (L-31). We follow simulation-reporting practice in pre-specifying and reporting the design, the number of replications, and how it was chosen (L-27), and adopt the standard separation of parameter, stochastic, heterogeneity and structural uncertainty (L-30).

Monthly revenue is generated multiplicatively from a baseline level with deterministic trend, a fixed seasonal multiplier, a shock multiplier and lognormal idiosyncratic noise. Operational variables are **derived through accounting identities rather than sampled independently** — `orders = R/AOV`, returns and fees as deductions — correcting an independence defect in the project's original generator. Each run applies all arms to the same path; 500 paths per scenario, base seed 20260803 **[baseline_v3_canonical.json → /n_paths, /base_seed]**.

6.1 Parameters and scenarios, in one place

All values are taken from the frozen specification and the registered artifacts. **Nothing here is recalibrated or recomputed.**

Item	Value	Source
Baseline monthly revenue R_0	185,000,000 VND	spec §4

Item	Value	Source
Advance `A`	185,000,000 VND	`baseline_v3_canonical.json` → `/terms/A`
Remittance rate `r`	0.10	`/terms/r`
Cap factor — illustrative	`f = 1.20` → cap 222,000,000 VND	`/terms/f`, `/terms/cap`
Cap factor — reference-path grid match	`f* = 1.0945` → cap 202,482,500 VND	`baseline_equalcost_v2_canonical.json` → `/terms`
Horizon `T`	24 months; recovery checkpoints at 12, 18, 24	spec §3
Paths per scenario	500	`/n_paths`
Base seed	20260803	`/base_seed`
Bootstrap seed	90210, independent stream	spec §10
Remittance basis	`net_sales` (GMV net of returns); burden displayed against GMV	spec A-1
Platform fee rate	0 (declared arbitrary-and-awaiting-justification)	spec
Idiosyncratic noise	`ε_t ~ LogNormal(−σ²/2, σ²)`, `E[ε] = 1`, `σ = 0.15`	spec §4
Seasonality	fixed 12-month multiplier, mean-normalised to 1.0; `moderate` = ±20%, `strong` = ±40% amplitude — **assumptions, not measurements**	spec §5.1
Trend	`g ∈ {−0.03, 0, +0.03}` monthly	spec §4
Benchmark **FIX-A**	matched principal, total repayment and term on the reference path; `N·P = C`	spec §7.1
Benchmark **FIX-B**	illustrative amortizing reference, `j = 18%` nominal, `N_B = 12` — an assumed input, not a market rate	spec §7.2, A-8

Scenarios. Ten non-closure scenarios, all at 500 paths: `stable` (flat seasonality, no growth, no shock); `seasonal` and `seasonal_strong` (±20% / ±40% amplitude); `growth` (+3% month-on-month); `gradual_decline` (linear decay to −40% over six months, then flat); `sustained_decline` (from month 7, −40% permanently); `severe_downturn` (−60% for six months, then six-month recovery); `disruption_1m` (one month at half revenue); `platform_outage` (one month at 30% of revenue, `d = 0.7`); `returns_spike` (return rate ×3 for three months, reducing net sales without reducing gross orders).

Closure scenarios, run at both cap factors: `closure_m7` and `closure_m13` (permanent cessation from month 7 / month 13); `temp_closure` (three zero months from month 7, then partial recovery to 50%).

Under-reporting sweep. `ω ∈ {1.00, 0.95, 0.90, 0.80, 0.70}`, applied mechanically to the observed base.

Two terminological commitments, both load-bearing.

First, intervals. Every interval reported in the underlying artifacts is a **Monte Carlo interval over simulated paths**. It measures whether enough paths were run for a number to be stable under our chosen generative parameters. It is **not** a confidence interval and says nothing about real sellers; running more paths narrows it without adding a single fact about the world. We could find no single source stating this prohibition, and we present it as synthesis: Monte Carlo error is a distinct quantity from inferential uncertainty (L-28); Monte Carlo standard error quantifies uncertainty arising from finite repetitions (L-27); and international measurement standards deliberately use "coverage interval" rather than "confidence interval" for precisely this distinction (L-29).

Second, convergence. We checked convergence for **two estimators on one scenario**: the paired differences `Δn_HPB(θ = 0.15)` and `ΔRR(12)` under a sustained −40% decline, which move **0.0027 months** and **0.042 percentage points** between 5,000 and 10,000 paths **[P-3 | `validation_v2_canonical.json → /convergence`]**. This does not establish that every estimator in the study is converged, and we do not claim it.

A disclosure about the burden denominator. The contract charges its share on **net sales**; the burden statistic we display uses **GMV** as its denominator. The identity `burden = r·(1 − return rate)` therefore holds only under three conditions, all of which must be stated: **(i)** before the final clipped payment, since the last payment is `min(r·B_t, remaining)` and can be lower than `r·B_t`; **(ii)** where net sales relate to GMV in that form, i.e. net sales = GMV × (1 − return rate); and **(iii)** subject to operational integer-VND rounding, which perturbs the ratio at the đồng level. Within those conditions the displayed burden is constant only where the net-sales/GMV ratio is fixed — most registered scenarios hold it fixed, and `returns_spike` is the explicit exception where it varies.

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7. Analytical propositions

This section is an **analytical verification layer**, not a set of claimed novel theorems. Most of what follows is an elementary consequence of the contract definition; the reason to state and prove it is to fix which results are properties of the **contract** and which are artefacts of the **simulation**. Each is proved in `research/DERIVATIONS.md` and asserted numerically against the engine, so the boundary is enforced rather than asserted.

These are not simulation output, and the simulated-magnitude qualifier that governs §8–§10 does not apply to them. Their limitation is different in kind: they describe a contract, and a contract is not a market.

P1 — Burden is definitional. Until the final capped payment, `p_t / B_t = r` exactly on the contractual base. The revenue-contingent arm's flat contractual burden is a definition, not an empirical finding, and no simulation could provide evidence for it.

P2 — Fixed burden has elasticity −1. For the fixed arm, burden `P/B_t` is strictly decreasing in `B_t` with constant elasticity −1: a revenue fall by factor `λ` raises burden by exactly `1/λ`. A −50% month doubles it, for any `P` and any `B_t`. This is the structural asymmetry the study exists to quantify, and it requires no calibration to state.

P3 — Cap condition. The cap is reached by month `k` iff `S_k ≥ A·f/r`. Duration is the **first passage time** to that threshold — a property of the cumulative trajectory, not of its endpoint. Two paths with the same terminal cumulative base need not cross in the same month.

P4 — Recovery ordering, exactly, and only on its domain. The result holds **for horizons `k ≤ N`** (the fixed arm's term) **and only before either contract has reached its cap**. On that domain, revenue-contingent cumulative recovery exceeds fixed cumulative recovery through `k` **iff** the realised mean eligible base `(1/k)·S_k` exceeds `B* = P/r`. Outside the domain — beyond `N`, or once either arm has capped out — the comparison is governed by the cap, not by this inequality, and the iff does not apply. This is not captured by the labels "declining" versus "non-declining": a declining path whose realised mean still clears `B*` leads, and a flat path below `B*` lags. A corollary matters for reading §8: benchmark matching sets `N = ⌈C/(r·B̄)⌉`, so whenever the term rounds up, `B* < B̄` strictly.

P5 — Under-reporting. If the provider observes `ω·B_t`, cumulative recovery scales exactly by ω and the required cumulative base rises by `1/ω`. **Duration does not scale by `1/ω`** — the threshold does, and duration is first passage to it. Only **uncapped** payments rescale;

the final payment is clipped to the remaining balance and need not scale. Invariance of the total is conditional on the cap still being reached.

P6 — Cost and rate. (a) `A·f` is the contractual repayment **target**, path-independent; realised total repayment equals it **only upon completion**. (b) Effective APR is not a well-defined property of the contract: **among paths whose payment stream admits an internal rate of return**, two sellers on identical terms face different APRs purely because revenue arrives at different speeds. Existence is a separate question from completion: a path that never reaches the contractual target still has a defined return on the payments it made, and the rate is undefined only where **no payment occurs at all**, so the equation has no root (A-9). Where the target is not reached, the figure is an **observed-window** rate over the horizon rather than the cost of a repaid contract, and must be read beside the incomplete-recovery rate.

P7 — Completion, exactly. The contract completes over horizon `H` **iff there exists a finite `t ≤ H` with `S_t ≥ 0`**, where `θ = f·A/r`. For a **finite** `H` this reduces to `S_H ≥ θ`, since `S_k` is non-decreasing. For an unbounded lifetime it does not reduce: the limit is not a partial sum. Writing `S_∞` for the lifetime sum, `S_∞ > θ` strictly implies completion, `S_∞ < θ` precludes it at any horizon, and `S_∞ = θ` completes only if a finite partial sum **attains** `θ` — which a strictly positive infinite series never does. Under geometric decline the condition is `ρ > ρ*` **strictly**, where $\rho^* = 1 - r \cdot B_0/(f \cdot A)$: **11/12 ≈ 0.9167 at `f = 1.20` and 0.9086 at `f* = 1.0945` [M-6]**. The threshold moves with the price, so 11/12 must never be quoted as *the* threshold.

Routes to incomplete recovery include zero revenue, a binding maturity or write-off, a binding evaluation horizon, and strictly positive but fast-decaying revenue whose lifetime sum converges below `θ`. **This list is illustrative, not exhaustive** — the complete characterisation is failure of the finite-time condition, and one route was already overlooked in an earlier draft of our own derivations.

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8. Results

Across the ten non-closure scenarios, all figures below are means across 500 simulated paths **[`baseline_v3_canonical.json → /scenarios`]**.

Stable scenario [`baseline_v3_canonical.json → /scenarios/stable`]. The revenue-contingent arm runs **12.86** months mean duration with mean displayed burden **0.0933** and recovers **96.56%** of its target by month 12. FIX-A runs 13 months, burden **0.0943**, RR(12) **92.31%**. FIX-B runs 12 months, burden **0.0936**, RR(12) **100%**.

The revenue-contingent arm leads FIX-A on twelve-month recovery by **4.25 percentage points** at exactly baseline revenue **[I-3 | `baseline_v3_canonical.json → /scenarios/stable/{RBF,FIX-A}/recovery_ratio/12`]**. **This is an artifact of integer rounding in the matching rule, not an economic finding** — P4's corollary: $C/(r \cdot \bar{B}) = 12.37$ rounds to 13, giving $B^* \approx 0.951 \cdot \bar{B}$. We report it as an artifact because reporting it as a result would be the kind of thing this paper argues against.

Severe downturn. The trade-off appears in full **[S-1 | `/scenarios/severe_downturn`]**:

	Duration (months)	Mean burden	Months above 15% burden	RR(12)
RBF	**18.718**	0.0943	**0.0**	**65.46%**
FIX-A	13.0	**0.1636**	**6.85**	**92.31%**
FIX-B	12.0	0.1606	5.906	100%

The revenue-contingent arm removes **6.85** months above the 15% threshold and holds mean burden essentially at its stable-scenario level — a rise of **1.14%** against stable — while the fixed arm's burden rises by **73.5% [S-1 | `baseline_v3_canonical.json → /scenarios/{stable,severe_downturn}/{RBF,FIX-A}/burden_mean, /n_high_burden/0.15`]**. The same mechanism costs the provider **26.85 percentage points** of twelve-month recovery and extends mean duration by **5.718 months**.

Neither column may be reported without the other. In this scenario the burden reduction and the recovery delay are one mechanism, and a chart showing only the first is a misrepresentation of the second. The pairing rule is general; the *direction* of the recovery effect is not, and is stated per scenario under P4.

The pattern is monotone in shock severity across the stress scenarios: gradual decline (RBF duration 16.148, RR(12) 84.21%; FIX-A 0.894 high-burden months), sustained decline (17.952, 76.04%; FIX-A 3.1), severe downturn as above **[`/scenarios/*`]**.

On the thresholds. The 10/15/20/25% bands are **illustrative reporting bands chosen for this study**. They are not validated hardship cutoffs, and no claim is made that crossing one causes distress **[Q-4]**. We report counts across bands rather than a single mean because burden distributions, not burden means, are what the repayment-burden literature identifies as decision-relevant (L-13).

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9. Pricing versus structure

At the illustrative cap factor `f = 1.20`, the simulated revenue-contingent contract is substantially more expensive than the illustrative 18%/12-month amortizing reference. At `f* = 1.0945` it is not.

What is and is not held constant. The **pre-cap payment rule** is identical in both cases: each payment is `r` times the remittance base, unchanged by the cap factor. What the cap factor changes is the **contractual target** `A·f`, and therefore the threshold `Θ = f·A/r` that the cumulative base must cross — so completion timing, the size of the final clipped payment, and the realised payment stream all move with price. Price and the payment rule are **analytically separable**; they are not **outcome-independent**. Both jointly determine what actually happens on a path.

It would therefore be wrong to say repricing leaves "structural behaviour unchanged". What it leaves unchanged is the rule; what it changes is when that rule stops applying. Whatever one concludes about *cost* is nonetheless a conclusion about the price, not about revenue-contingent repayment as a structure.

`f* = 1.0945` is the **nearest match on the swept cap-factor grid** to the reference path's effective APR, not an exact solution: **19.537656%** against **19.561817%**, a residual of approximately **0.02416 percentage points [P-1 | `validation_v2_canonical.json → /pricing/equal_cost, /pricing/benchmark_b_apr`]**. An exact match is not generally attainable, because duration is integer-valued and the achievable APRs therefore form a discrete set. We report the residual rather than smoothing it away.

Repricing under the identical payment rule **[`baseline_equalcost_v2_canonical.json → /scenarios`]** shortens duration in every scenario — stable **11.784**, sustained decline **16.01**, severe downturn **17.504** — precisely because a lower cap is a nearer threshold. This is the point: the payment rule did not move, the stopping condition did, and the realised outcomes moved with it.

9.1 A retraction, reported as part of the result

An earlier version of this project stated that "RBF costs approximately 2.3× the interest of a conventional loan." **That claim is withdrawn**, and the reason is the argument of this section. It was wrong on two counts: it fixed `f = 1.20` as though it were intrinsic, when P6(a) shows the **contractual repayment target `A·f` is proportional to `f`** — and realised repayment equals that target **only when the contract completes**, so even the target is not a realised cost on every path; and it quoted a single effective APR as though it were a contract property, when P6(b) shows that **among paths admitting an internal rate of return** the APR is jointly determined by the price and the revenue path. Non-completing paths are not excluded from that set: `closure_m7` reaches the target on no path and still has a defined mean rate of **−86.5129% [S-3 | `baseline_closure_v2_canonical.json` → `/scenarios/closure_m7/RBF/apr_mean`]**.

We report the retraction rather than quietly correcting it because it is the concrete instance of the conflation this paper argues against, produced by this project while actively trying to avoid it. That it survived several earlier drafts is the useful part of the anecdote.

Two external observations support treating price and structure separately as a matter of course.

First, legislators and regulators have concluded that factor-rate pricing is not comparable to APR without conversion. California's SB 362 (2025) lists among its legislative findings the practice of "describing the price of credit as 'X% fee rate' or 'Y% factor rate,' particularly when those 'rates' diverge materially from the APR" (L-45, SECTION 1(e)(3)), and requires a provider to state the APR whenever it states a charge or pricing metric after extending a specific offer (L-45, Financial Code §22806). California's earlier 2018 statute and its 2022 implementing regulations established the underlying APR-disclosure requirement (L-24), and New York's 23 NYCRR 600 imposes standardised commercial-financing disclosure including APR (L-23). Whether such state regimes are preempted by federal Truth in Lending has itself been the subject of a formal federal determination (L-25), which indicates the area is contested rather than settled. These are determinations by a legislature and by regulators — evidence that the comparability problem is recognised in law, not quantified research findings.

Second, contingency **can** command a premium where it transfers risk to the financier or creates adverse selection. This has been demonstrated in **human-capital financing**, where adverse selection implies a typical borrower would repay roughly $1.64 in present value per dollar financed for an equity-like market to clear (L-18), and in **sovereign** state-contingent instruments, which carry persistent premia (L-19). **Neither result establishes a universal rule.** We do not claim that contingency *must* be priced above fixed debt, nor that a parity-priced revenue-contingent contract is anomalous: those are claims about SME revenue-based financing, and the cited evidence is from other domains. **Whether a pricing premium applies to SME RBF, and how large it would be, is an open empirical question** that the literature reviewed does not answer.

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10. Closure, incomplete recovery and censoring

10.1 The null result, and its boundaries

Across the ten non-closure scenarios, incomplete recovery is **0.0%** and total repaid is identical at the cap **[S-2 | `baseline_v3_canonical.json` → `/scenarios/*/RBF/incomplete_recovery_rate`, `/total_repaid_mean`]**.

This is **horizon- and scenario-bounded**, and stating it alone would be misleading: none of those ten scenarios reaches zero revenue, and the horizon is 24 months against a 13-month matched base term. An earlier version of this project reported this null as evidence that provider exposure was "duration risk, not principal loss". That framing is withdrawn.

10.2 Where recovery actually fails

Permanent closure occurring **before completion, while a contractual balance remains outstanding**, is the case where recovery genuinely fails. Both halves of the table below carry their own source: the `f = 1.20` columns are **[S-3 | `baseline_closure_v2_canonical.json` → `/scenarios/*/RBF/incomplete_recovery_rate`, `/recovery_ratio/24`]** and the `f* = 1.0945` columns are **[S-4 | `baseline_closure_equalcost_v2_canonical.json` → same paths]**.

Scenario	Incomplete recovery, `f = 1.20`	RR(24)	Incomplete recovery, `f* = 1.0945`	RR(24)
`closure_m7`	**100.0%**	44.30%	**100.0%**	48.57%
`closure_m13`	**76.2%**	96.53%	**7.6%**	99.88%
`temp_closure`	2.0%	99.98%	**0.0%**	100%

Three things in this table deserve emphasis.

Timing decides it, not zero revenue as such. A temporary three-month cessation leaves only 2.0% of paths incomplete at `f = 1.20` and none at `f*`. Closure at month 7 — before the matched 13-month term — leaves every path incomplete. It is not the presence of zero-revenue months that prevents completion; it is permanent cessation before the threshold is crossed while a balance remains.

Price moves the failure rate by an order of magnitude. `closure_m13` incomplete recovery falls from 76.2% to 7.6% on the cap factor alone **[S-4 | `baseline_closure_equalcost_v2_canonical.json`]**. This is P6(a) made visible: a lower cap is a nearer threshold.

Incomplete recovery is not principal loss. `closure_m13` recovers approximately **214.3M VND** against a **185M** advance — the principal is covered despite 76.2% of paths failing to reach the contractual target. Only `closure_m7` shows a principal shortfall, recovering approximately **98.3M**, and it recovers the **same absolute amount at both cap factors** because that path is revenue-limited rather than cap-limited **[I-3 | recovered amount computed as `/scenarios/*/RBF/recovery_ratio/24 × /terms/cap`, from `baseline_closure_v2_canonical.json` and `baseline_closure_equalcost_v2_canonical.json`; advance from `/terms/A`]**. A 76.2% incomplete-recovery rate is not a 76.2% loss rate, and we are careful not to let the first number stand in for the second.

10.3 Censoring, and what our own statistics omit

Mean duration and mean effective APR are conditioned, but **not on the same event**, and an earlier version of this paper described them as though they were.

Mean duration is a survivor statistic. It is computed only over paths that reached the repayment target within the window, so it estimates `E[T | completion occurs by horizon H]`, not `E[T]`, and it describes the contracts that finished rather than the portfolio. (We avoid writing the conditioning event as `T ≤ C`: in this paper `C` already denotes the VND contractual cap, and reusing it as a censoring time would be ambiguous.)

Mean effective APR is conditioned on the existence of an internal rate of return, which is a different and generally larger set. A path can miss the contractual target and still have a well-defined return on the payments it did make; the rate is undefined only where no payment occurs at all, so the equation has no root (A-9). In `closure_m13` at `f = 1.20`, **119 of 500** paths complete while **all 500** have a defined rate — 381 paths are simultaneously incomplete and rate-defined **[`baseline_closure_v2_canonical.json → /scenarios/closure_m13/RBF/completed_count, /apr_defined_count`]**. Averaging over the rate-defined set gives **30.33%**; averaging over the completed set alone gives **39.37%**. Reporting the first under the second's label understated the completed-path figure by roughly nine percentage points. The artifacts now publish `completed_count`, `completed_rate`, `apr_defined_count` and `apr_defined_rate` so that neither denominator has to be inferred.

Where a rate is reported for an incomplete, non-absorbing path it is an observed-window figure — the return on payments made within the 24-month horizon, not a final lifetime return — and it must be read beside the incomplete-recovery rate.

Timing here is a cost in the sense the fixed-income literature gives it — duration as the weighted-average timing of cash flows (L-20) — and lender-side recovery ratios have been compared across contingent and fixed scheme designs before, though in student lending rather than business finance (L-22). The censoring framing is standard sample selection: computing a statistic on a non-randomly selected subsample is a specification error (L-36), and conditioning on completion is conditioning on a common effect (L-37). The machinery is standard right-censoring (L-32, L-33), and in an economics setting the duration-data treatment (L-34, L-35) is the appropriate reference. We found no dedicated source quantifying this particular bias, and we do not cite one; the direction follows immediately from the definition.

The practical consequence is a reading rule. **A scenario with a short mean duration and a high incomplete-recovery rate is not a fast contract — it is one where the slow paths were dropped rather than counted.** `closure_m13` at `f = 1.20` reports a mean duration of **11.99** months alongside **76.2%** incomplete recovery **[`baseline_closure_v2_canonical.json → /scenarios/closure_m13/RBF/duration_mean, /incomplete_recovery_rate`]**; reporting the first without the second would invert the finding.

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11. Product implications

These are judgements we draw, not measurements.

Report price and structure separately, always. Otherwise a pricing choice is silently attributed to a structural property. This is the methodological recommendation, and §9.1 is the evidence that it is easy to get wrong.

Provider exposure has two distinct forms. Duration risk in ordinary downturns, and an unrecovered contractual balance where permanent closure precedes completion. Both belong in a term sheet, and the second cannot be inferred from ten scenarios that never reach zero revenue.

The guardrail result, both halves. The design includes a hardship payment floor and a payment ceiling. The **floor never activates on any path — 0 of 36,000** month-observations — because the floor multiplier is below the hardship threshold (`μ = 0.25 < h = 0.50`), making it dead by construction **[`validation_v2_canonical.json → /rbf_g_breakpoint/pmin0.25_hard0.5/floor_months, /total`; registry null N-2′]**. The **ceiling does bind**: 6,009 of 36,000 observations in the breakpoint scan, and in the baseline it changes results in **6 of 10** scenarios — mean APR in 6, mean burden in 6, recovery ratio in 3, mean duration in 1 **[`baseline_v3_canonical.json → /scenarios/*/RBF-G`]**. The differences fall below display precision, which is why they went unnoticed through several reviews. **Invisible at display precision is not identical**, and presenting the floor null alone would read as a whole-arm null, which is false. As specified, the floor is decoration; the ceiling is not.

No predictive-validity claim, and a worked reason. The demonstration product includes a machine-learning risk score. It is trained on synthetic data whose default label was generated by a hand-written weighted formula over the same features the model consumes — so the model was scored on its ability to rediscover that formula. Our own reproducible evidence: the AUC of the generating function against its own label is **0.9098**, against the reported ensemble AUC of **0.9182 [R-000 | `research/analysis/00_audit_evidence.py`]**. The reported figure measured the chosen noise variance, not predictive skill, and is **withdrawn**; we cite it here only to explain the withdrawal. The general mechanism — leakage inflating apparent performance — is documented across 294 papers in 17 fields (L-38). We found no source stating the specific circularity result, and argue it from our own evidence rather than attaching a citation that does not establish it.

The fixed arm is an optimistic benchmark. Fixed payments are modelled as made in full and on time in every month of the schedule. The fixed arm therefore represents scheduled recovery under that assumption, not realised recovery, and the comparison flatters it. Correcting this would require a default model, which this study does not have and does not claim to have.

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12. Limitations and responsible use

No observed data. No observed seller revenue, repayment or default outcome exists anywhere in this study. **Every reported simulated magnitude is simulation output**; the analytical propositions of §7 are proved results, not simulation output, and are limited differently — they describe a contract, and a contract is not a market. Nothing here is evidence about Vietnamese sellers, or about any seller.

No causal, predictive or population claim. The design supports statements about contract mechanics under our assumptions. It does not support statements about what sellers would do, which sellers would accept these terms, or how a portfolio would perform.

No affordability claim. Burden is measured against **revenue**, not against what the seller retains. Margins, operating costs, reserves and other obligations are outside the model, so a lower burden here does not establish that a contract is affordable **[Q-2]**. The threshold bands are reporting devices, not validated hardship cutoffs **[Q-4]**.

No default claim of any kind, in either direction. This study contains no default model, no observed default event and no credit-loss model. §10 does not show defaults; it shows **failure to reach the contractual repayment target within the stated horizon**. What §10 establishes is narrow and sufficient: **any guarantee of contractual completion is disproved.** Revenue-contingency changes who bears timing risk; it does not remove risk; and nothing here licenses a statement about default rates.

Unsourced parameters. The benchmark rate `j = 18%`, term `N_B = 12`, the seasonality shapes, the margin `m` and fixed cost `F` are assumptions. None was externally sourced **[Q-5]**. The generalization limit binding on all outputs is that findings are statements about contract mechanics in a Vietnam-motivated, illustratively parameterised range (A-8).

Synthetic-data limits. Using synthetic data as though it were real yields analyses that do not generalise (L-39); institutional guidance identifies nuances easily overlooked in deployment (L-40). Our synthetic component is a demonstration, and §11 states what it cannot support.

Self-reported-data limits. The under-reporting sweep is motivated by the finding that evasion is near zero for third-party-reported income and substantial for self-reported income (L-42), and that the self-employed under-report by roughly 25% in household surveys (L-43). Neither magnitude transfers to marketplace revenue, and we do not transfer them. Measurement error in self-reported data is frequently non-classical, biasing in unpredictable directions rather than merely attenuating (L-41); the broader evasion literature identifies information reporting as the mechanism that makes the difference (L-44). **Our sweep is mechanical**: we impose ω and observe the contract's response. Russel et al. (2025) document *behavioural* revenue diversion under real contracts (L-07) — a materially different and harder object, which we do not model.

What uncertainty we do and do not report. Against the standard taxonomy (L-30), this study reports: **stochastic variation conditional on the chosen parameters** (across 500 simulated paths per scenario); **selected deterministic sensitivity analyses** (the cap-factor sweep, the ω sweep, the closure scenarios); and **limited convergence evidence for two estimators in one scenario** (§6). It reports **no quantified parameter uncertainty** — no parameter was assigned an uncertainty distribution and propagated, so we must not describe the study as reporting parameter uncertainty — and **no quantified structural uncertainty**, which is to say we have not quantified the consequences of the revenue process itself being the wrong model. By that taxonomy, structural uncertainty is the largest unquantified component.

Responsible use. This is not underwriting guidance. The demonstration product must not be presented as a validated risk model, and its risk score must not be described as a probability of default. Misrepresentation of terms in this product class has been the subject of federal enforcement (L-26), which is a reason for care rather than a finding about any particular provider.

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13. Conclusion

Revenue-contingent repayment and fixed instalment repayment differ in *when* money moves, and that difference has two faces.

In the simulated scenarios examined here, the revenue-contingent contract holds the seller's **displayed** payment burden roughly flat under adverse revenue paths — a statement that holds for scenarios in which the net-sales/GMV ratio is fixed, and for periods before the final clipped payment — while the fixed contract's burden rises in inverse proportion to revenue. The same mechanism moves the provider's recovery, and where permanent closure precedes completion it leaves a contractual balance unrecovered.

In which direction it moves recovery is scenario-dependent, not uniform. It follows the exact P4 condition on realised mean eligible base against `B* = P/r`, and both directions occur in this scenario library: recovery lags in the severe-downturn scenario and leads at exactly baseline revenue. A summary that states a direction without naming a scenario is not supported by these results. One further qualification travels with the sentence above: **the fixed arm is an optimistic scheduled-recovery benchmark**, since every instalment is assumed paid in full and on time. Reporting either face alone misstates the contract.

The price of the contract is a separate question from its payment rule, and the two are easy to conflate — an earlier project draft stated the conflation, and it was withdrawn before any external publication. At one cap factor the revenue-contingent arm looks expensive; at another it does not; the structure is identical in both. Any cost comparison that does not name its cap factor is uninterpretable.

What would change these conclusions, and what would not. Observed seller revenue paired with adjudicated repayment outcomes would let us **calibrate the revenue process** — replacing assumed seasonality, volatility and shock shapes with measured ones — and would let us check whether the simulated magnitudes resemble anything real. That would be a large improvement.

It would **not**, on its own, deliver the comparison this paper makes. Observational data cannot show both contracts on the **same seller under the same realised revenue path**, because each seller takes at most one contract. Recovering that counterfactual would still require randomisation across contract types, a credible quasi-experimental design, or a structural model with an explicit counterfactual. Calibration and causal identification are different problems, and observed data solves only the first.

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14. References

All entries verified against a publisher deposit, DOI resolution, or the issuing institution's page. Full annotation, including what each source does **not** support, is in `research/publication/LITERATURE_MATRIX.md`.

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15. Reproducibility statement

Artifacts. Every **simulation-result table and figure** in this manuscript derives from one of five canonical, checksummed artifacts. Analytical results (§7) are derivation-backed, not artifact-backed, and external facts are literature-backed; neither is covered by this table.

Artifact	SHA-256
`baseline_v3_canonical.json`	`818c145ad557ea1f95311fe80d311252 103464ba7a7ecac602aab67374ae8308`
`baseline_equalcost_v2_canonical.json`	`9cc6885a3d0d2d54fb08ae85301ae588 9e7059f2780cdcfca693b3a8ec47802d`
`baseline_closure_v2_canonical.json`	`c032625a8e7c17c55a590eac673e447f 178fdb192812fe98ee6df0b6e228fd75`
`baseline_closure_equalcost_v2_canonical.json`	`de7de916cfa73b7ff1c3b153f068ecda e90670ad4e0283e27c9ce36bb544458a`
`validation_v2_canonical.json`	`ba79342ef2865a8a439f3f1a22a94819 52c459b96e11f88e8d3be3384fd5b682`

Superseded by A-9 (D-049), preserved byte-for-byte. Every figure published before 2026-08-20 was computed from these files, so the record of what was published stays independently verifiable. They are historical evidence: never cited as current, never re-generated, never deleted. Their integrity is asserted by `backend/tests/test_validation_artifact.py`.

Superseded artifact	SHA-256
`baseline_v2_canonical.json`	`264d319be6854533b4a51a7114c34dbf fb0728d9ed3bfd50973b6ab4ac5a7849`
`baseline_equalcost_v1_canonical.json`	`6f9c71b111400aea1b2ea5c06527404f 849fa390de0eef47e22b75a552da68e7`
`baseline_closure_v1_canonical.json`	`0fe503d7f96b4c21d68b2fb812e0e964 5ac21bdb6308208be4182cdef6179470`
`baseline_closure_equalcost_v1_canonical.json`	`49b6f8ef19c81eebe1288d0d090c858a 64b30f35cd5263afd6f4a971696b15f9`

Superseded artifact	SHA-256
`validation_v1_canonical.json`	`f89fd2baab7f5628f9aed7e7a6be7ae4 0e0e72919b0f9f41e20875946c67ddb4`

Reproducibility, stated at the strength the measurement supports. All five artifacts reproduce **numerically at published precision** from a clean tree on every platform tested. **Byte equality is platform-dependent and is not claimed across platforms:** on Linux/aarch64 CPython 3.10.12 all five regenerate byte-identically; on macOS CPython 3.11.5, `baseline_v3` differs in 9 last-bit floating-point values and `baseline_equalcost_v2` in 2, while the other three are byte-identical. Verify with `python3 research/verify_reproduction.py`, which regenerates into a scratch tree and reports byte equality and numeric-leaf equality as separate columns.

An earlier version of this project claimed byte-for-byte reproducibility without qualification, on the basis of an evidence step that re-hashed the committed file rather than regenerating it. That step could not have failed for any reason connected to determinism. Both the claim and the evidence are corrected.

A resolved inconsistency, and where it survives. The five **superseded** artifacts each embed a metadata field `canonical.determinism` carrying an unqualified byte-identity claim that D-041 withdrew. They were deliberately not rewritten: correcting a sentence *about* reproducibility by regenerating the files whose checksums record it would have destroyed the evidence in order to tidy it. That trade was reopened only when A-9 demonstrated a genuine defect in the rate layer, at which point regeneration had to happen anyway — so the **current** artifacts carry the corrected wording: numeric equality at published precision across tested platforms, byte equality only within a fixed runtime. The superseded files keep the old string, are preserved byte-for-byte as the record of what was published, and a regression test asserts that no public surface renders the field (D-043, D-044, D-049).

Tests. 1,080 non-browser tests pass: 437 backend and 643 simulation. Nine browser checks are defined and **excluded from that total**. They passed in the earlier browser-capable run recorded at D-036. In environments lacking Playwright or Chromium they skip; pytest may report one skipped module or nine skipped cases depending on what is installed. Skips are never counted as passes.

Governing documents. `research/METHODOLOGY_SPEC.md` v1.0 + amendments A-1…A-9 (frozen specification); `research/DERIVATIONS.md` (propositions and proofs); `research/CLAIM_LEDGER.md` (what may be claimed, with required qualifiers); `research/RESULTS_REGISTRY.md` (every registered result); `research/DECISION_LOG.md` (every decision and supersession, including our own retracted claims).

Appendix A. Source-note convention

Quantitative statements in this manuscript are of four different kinds, and a single note format cannot serve all four. An earlier draft promised one that it could not satisfy for external statistics or analytical constants. The rule adopted here is type-appropriate:

Claim type	Note format
External fact or finding	**(L-nn)** — entry in `research/publication/LITERATURE_MATRIX.md`
Analytical claim or constant	proposition reference in `research/DERIVATIONS.md`, and a ledger **M-n** where one exists
Simulated magnitude	**[artifact → JSON path]**, plus a ledger ID (S-, P-, I-) where one exists
Test count, checksum, reproducibility	reproducibility evidence, §15

A quantitative statement without the note appropriate to its type is a defect, not a stylistic choice. **Ledger IDs are never invented to fill the slot** — where no ledger row exists, the artifact path stands alone.

A reader wanting to check any figure in this paper can therefore go directly to its source: literature entries resolve to `LITERATURE_MATRIX.md`, analytical constants to the proof in `DERIVATIONS.md`, and every simulated magnitude to a JSON path inside one of the five checksummed artifacts listed in §15.